

Q- Compare the router-on-a-stick method you just configured with inter-VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

Method	Advantages	Disadvantages
Router-on-a-Stick	1. Low cost — uses a normal Layer 2 switch with a router. 2. Easy to configure for a small network using router sub-interfaces.	1. Limited performance — all inter-VLAN traffic passes through one router link. 2. Single point of failure — if the router or trunk link fails, inter-VLAN communication stops.
Layer 3 Switch	1. High performance — routing is performed directly inside the switch using hardware. 2. Better scalability — suitable for larger networks with many VLANs and high traffic.	1. More expensive than using a basic Layer 2 switch with a router. 2. More complex configuration — requires Layer 3 switch features such as SVIs and IP routing.